

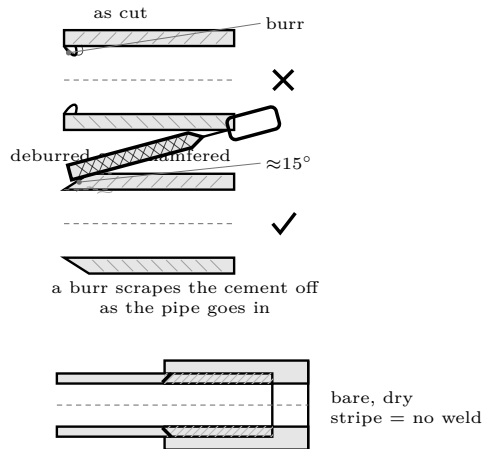
Craft 3: Deburr & Chamfer

Two minutes that decide whether the joint is real

Shop skill — transferable to anything you ever build

THE STANDARD

No burr anywhere, inside or out, and the outside edge of every pipe end carries a **small even chamfer** — roughly 1 to 2 mm at about 15 degrees, all the way round. Run a fingertip over the end: it should feel smooth in both directions and you should feel the chamfer as a gentle ramp, not a sharp step.



Why the outside burr matters

It is a ridge of raised plastic exactly where the joint needs contact. As the pipe enters the socket the ridge acts like a scraper and wipes the cement off the leading surface, leaving a dry stripe down the joint. From outside the joint looks perfect.

Why the inside burr matters

Inside a barrel it scores every projectile and disturbs the gas seal. Inside a chamber it collects residue. Neither is dramatic, and both are permanent.

Why the chamfer matters

Without it, the leading edge of the pipe meets the socket taper as a sharp square corner and plows the cement ahead of it instead of sliding through it. The chamfer lets the pipe enter cleanly and keeps the cement where it belongs.

Two minutes

This is the cheapest quality step in the whole build and the one most often skipped, because the pipe looks finished as soon as it is cut.

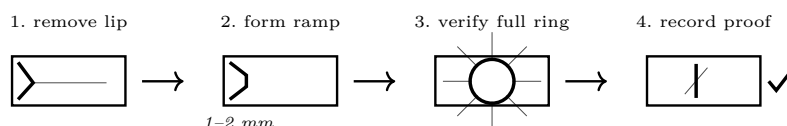
Tools

A sharp knife works. A half-round file is better. A dedicated deburring tool is a few pounds and makes it trivial. All three are fine; doing nothing is not.

Method

1. Remove the raised inside lip with a shallow tool angle. **Check:** a fingertip moving out of the bore does not catch. If it catches, make one more light pass and retest; do not cut an inside bevel.
2. Set the file at about 15 degrees to the outside surface. **Check:** the tool touches only the sharp corner. If it lies across the cut face, reduce the angle before removing material.
3. Rotate the pipe through one full turn while holding the tool angle steady. **Check:** a continuous narrow graphite-like highlight appears around the edge. If a bright square patch remains, work only that patch.
4. Measure the chamfer at its narrowest and widest points. **Check:** both are 1–2 mm. If either is under 1 mm, correct locally; if either exceeds 2 mm, remake the end.
5. Feel the inside and outside edges in both directions. **Check:** no burr catches. If one catches, correct it and repeat Steps 4–5.
6. Wipe away all swarf. **Check:** a clean white cloth returns without loose plastic. If debris remains, wipe again before recording acceptance.

Make the evidence in order



Ask at each arrow: Is the inner lip gone? Is the outside ramp only 1–2 mm? Does its width stay constant through one

full turn? Is the end clean and smooth in both fingertip directions? Record the measured chamfer width at its narrowest and widest points: _____ mm / _____ mm.

ACCEPT, CORRECT, OR REMAKE

Accept only when no burr catches, the full circumference carries the small ramp, and loose swarf is gone. Correct a remaining bright patch with a few controlled strokes, then repeat every check. Remake the end if the chamfer exceeds 2 mm, is deeply gouged, or the pipe end is no longer square. Do not hide a failed end in a socket.

What each defect looks like later

If you skip	What you get
Outside deburr	A dry stripe in the weld you cannot see and cannot test. The joint holds — until the day it does not.
Chamfer	Cement scraped into a ring at the socket shoulder and starved along the joint face. Also a much harder push, which people mistake for a tight fit.
Inside deburr	Scored projectiles, a poorer gas seal, and a ledge that collects residue.
Cleaning up swarf	Loose plastic in the chamber, and grit in the cement on the next joint.

Check it yourself. Close your eyes and run a fingertip all the way round the inside edge and then the outside edge. Eyes closed genuinely helps — your fingertip resolves a burr far better than your eye does, and closing your eyes stops you believing what the pipe looks like instead of what it feels like.

You should see: Final proof before assembly: show a second person the unbroken chamfer ring under raking light; let them repeat the two-direction fingertip check; point to the narrowest and widest measurements above. Both people initial here only after all three agree: maker _____ checker _____.

The habit worth taking away

The defects that matter most are usually the ones that become invisible the moment you finish the next step. Once a pipe is in a socket you will never see that burr again — so the check has to happen *before* assembly, every time, on every end. Build the sequence so that inspection comes before the step that hides the thing being inspected.

Inspection record

Pipe end	Inside lip gone	Ramp 1–2 mm	Even full turn	Clean / dry
End A	☐	_____ mm	☐	☐
End B	☐	_____ mm	☐	☐
End C	☐	_____ mm	☐	☐
End D	☐	_____ mm	☐	☐

Read the evidence

Circle the first failing observation, not the hoped-for result:

feel

→

raking light

→

measure

A smooth feel cannot prove width; a good measurement cannot prove the full turn. The three checks answer different questions.

Disposition

For each corrected end, record what changed:

Defect: _____

Correction: _____

Retest result: _____

☐ accept ☐ remake

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